StudyVerse AI – An AI-Based Interactive Educational Gaming Platform

A PROJECT REPORT

CS23533 - Foundations of Artificial Intelligence

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BONAFIDE CERTIFICATE

Certified that this project report “STUDYVERSE AI – AN AI-BASED INTERACTIVE EDUCATIONAL GAMING PLATFORM” is the bonafide work of “MADHUMITHA P (2116240701297) and TAMIZHARASI S (2116240701552)” who carried out the project work under my supervision.

SIGNATURE OF THE FACULTY INCHARGE

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SIGNATURE OF THE INTERNAL EXAMINER

ABSTRACT

Students often find traditional learning methods less interactive, making it difficult to maintain motivation and track their academic progress. Conventional educational platforms may lack engaging game-based activities, adaptive difficulty, and reward systems that encourage consistent learning.

This project presents **StudyVerse AI**, an AI-based educational gaming platform that provides interactive learning activities across five subjects: Mathematics, Science, English, Computer Science, and General Knowledge. The system uses rule-based adaptive difficulty to adjust question levels according to student performance and heuristic evaluation to assess learning progress. It also includes an optional AI Battle mode that uses the Minimax algorithm and Alpha-Beta Pruning for competitive decision-making. The platform is developed using React, Node.js, Express.js, and PostgreSQL. Gamification features such as XP points, levels, streaks, and badges motivate students to learn consistently. The system tracks quiz accuracy, subject progress, and achievements to provide an engaging and structured learning experience. It uses classical Artificial Intelligence techniques and rule-based decision-making rather than machine learning or model training.

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CHAPTER 1

INTRODUCTION

1.1 Background

Students need regular practice to improve their knowledge and track academic progress across different subjects. Traditional learning methods may lack interactive activities, making it difficult to maintain motivation and identify areas for improvement. StudyVerse AI addresses these challenges through an educational gaming platform that combines quizzes, rule-based adaptive difficulty, and gamification. It uses classical AI techniques, including heuristic evaluation and the Minimax algorithm with Alpha-Beta Pruning in AI Battle mode. Features such as XP, levels, streaks, and badges encourage students to learn consistently.

1.2 Problem Statement

To design and implement an interactive educational gaming platform that uses rule-based adaptive difficulty, classical AI techniques, and gamification to improve student engagement, track learning progress, and encourage consistent practice across multiple subjects.

1.3 Objectives

- To develop an interactive educational gaming platform covering five subjects: Mathematics, Science, English, Computer Science, and General Knowledge.
- To implement rule-based adaptive difficulty that adjusts question levels according to student performance.
- To use heuristic evaluation to assess performance and support adaptive learning.
- To implement an optional AI Battle mode using the Minimax algorithm and Alpha-Beta Pruning.

- To provide gamification features such as XP points, levels, streaks, and badges to motivate students.
- To track quiz scores, accuracy, subject progress, achievements, and overall learning performance.

1.4 Scope and Limitations

The StudyVerse AI platform provides educational games across five subjects: Mathematics, Science, English, Computer Science, and General Knowledge. It includes quizzes with different difficulty levels, rule-based adaptive difficulty, performance tracking, and gamification features such as XP, levels, streaks, and badges. The optional AI Battle mode uses Minimax and Alpha-Beta Pruning for decision-making. The system relies on predefined questions and rule-based logic rather than machine learning, so its adaptability is limited to the implemented rules and available question bank.

CHAPTER 2

LITERATURE REVIEW

2.1 Artificial Intelligence Techniques

Artificial Intelligence (AI) enables computer systems to make decisions and solve problems intelligently. StudyVerse AI uses classical AI techniques, including rule-based adaptive difficulty and heuristic evaluation, to support interactive learning. Rules adjust question difficulty based on student performance, while heuristic evaluation considers factors such as accuracy and recent performance. The optional AI Battle mode uses the Minimax algorithm with Alpha-Beta Pruning to improve decision-making efficiency. These techniques provide adaptive gameplay without requiring machine learning or model training

2.2 Rule-Based Reasoning and Decision-Making

Rule-based systems use predefined conditions and actions to make decisions based on available information. StudyVerse AI applies this approach to adjust question difficulty according to student performance. For example, two consecutive correct answers increase the difficulty, while two consecutive incorrect answers decrease it. Heuristic evaluation considers factors such as accuracy, response time, and recent performance. The optional AI Battle mode uses Minimax and Alpha-Beta Pruning to select competitive moves efficiently.

2.3 Rule-Based Expert Systems

Rule-based expert systems use predefined rules and stored knowledge to solve problems and make decisions. StudyVerse AI applies this approach to adapt quiz difficulty based on student performance. The system evaluates consecutive correct and incorrect answers to adjust difficulty levels. By combining rules with heuristic evaluation, it provides a structured learning experience without requiring machine learning or model training.

2.4 Educational Gaming Systems

Traditional educational systems often rely on fixed quizzes and exercises, which may offer limited engagement and adaptability. Gamified learning platforms improve motivation through rewards, levels, and achievements. StudyVerse AI combines educational quizzes with XP points, streaks, badges, and progress tracking. Unlike systems with fixed difficulty levels, it uses rule-based adaptation to adjust question difficulty based on student performance and encourage continuous learning.

2.5 Research Gap

Existing educational platforms provide quizzes and learning activities, but some offer limited adaptability and motivation. Traditional quiz systems often use fixed difficulty levels and provide limited opportunities to track learning progress. Gamification improves engagement, while Artificial Intelligence techniques can support adaptive decision-making.

The following gaps were identified:

- **Fixed difficulty levels:** Traditional quizzes may not adjust question difficulty according to student performance.
- **Limited engagement:** Repetitive learning activities may reduce student motivation.
- **Insufficient progress tracking:** Students need a convenient way to monitor scores, accuracy, subject progress, and achievements.
- **Lack of adaptive learning:** Some platforms do not respond to recent performance when selecting question difficulty.
- **Limited AI-based gameplay:** Educational games may not include competitive decision-making using classical AI algorithms.

- **Lack of motivation features:** Without rewards, levels, and achievements, students may find it difficult to maintain consistent practice.

StudyVerse AI addresses these gaps by combining educational quizzes, rule-based adaptive difficulty, heuristic evaluation, and gamification features. Its optional AI Battle mode uses the Minimax algorithm and Alpha-Beta Pruning for decision-making. The platform also tracks student performance through XP points, levels, streaks, badges, and subject progress. By integrating these features, StudyVerse AI aims to provide an interactive and engaging learning experience using classical AI techniques without requiring machine learning or model training.

CHAPTER 3

PROPOSED SYSTEM

3.1 ARCHITECTURE DIAGRAM

The system is developed using React, Node.js, Express.js, and PostgreSQL, and its architecture illustrates the interaction between the user interface, backend services, and database. Students interact with the presentation layer to access quizzes, educational games, and progress reports. The backend processes requests, applies rule-based adaptive difficulty, and uses classical AI techniques for decision-making. The database stores user details, questions, scores, XP, and achievements. The system tracks student performance and displays results, progress, and rewards through an interactive interface. The optional AI Battle mode uses Minimax and Alpha-Beta Pruning to support competitive decision-making.

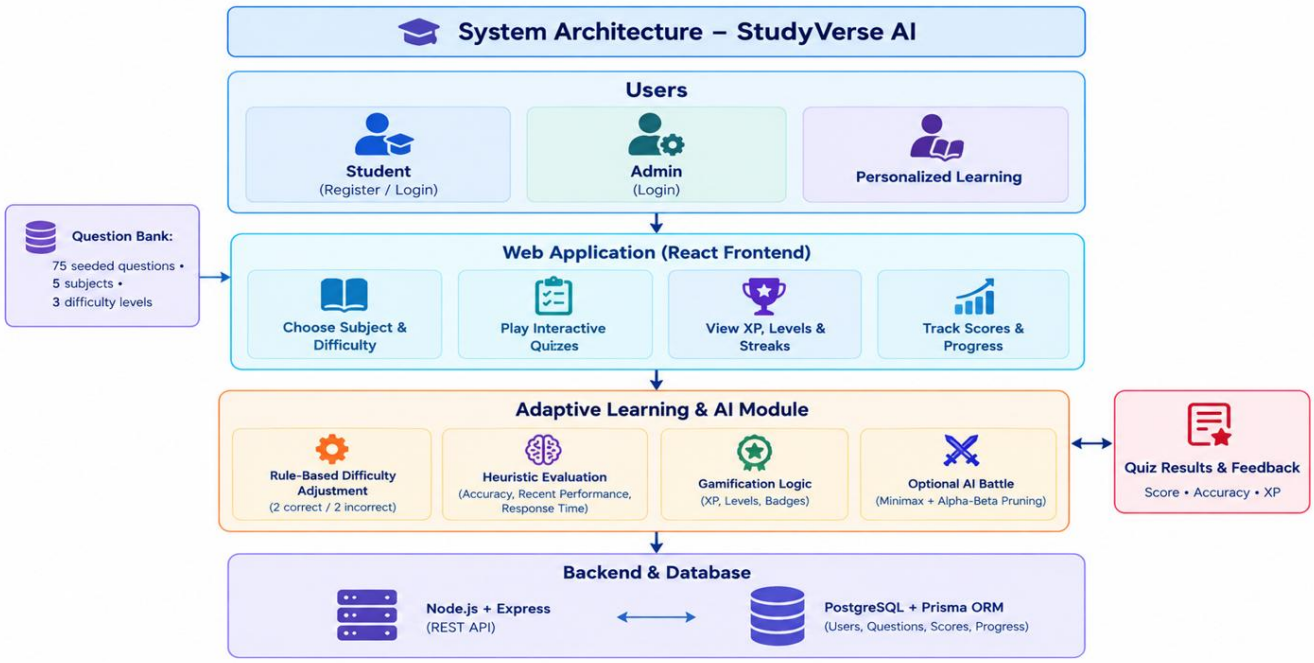


Figure 3.1 Architecture Diagram of StudyVerse AI

3.2 Working Principle

The student logs in and selects a subject, quiz, or game. The backend processes the request, retrieves questions from the database, and applies rule-based logic to adjust difficulty based on student performance. The system evaluates answers, calculates scores and XP, and updates progress, levels, streaks, and badges. In the optional AI Battle mode, the Minimax algorithm with Alpha-Beta Pruning supports decision-making. Finally, the interface displays the results, rewards, and updated learning progress to the student.

3.3 Knowledge Representation

StudyVerse AI stores user information, questions, quiz results, and learning progress in a PostgreSQL database managed using Prisma ORM. The system uses predefined rules to adjust question difficulty and heuristic factors to evaluate student performance. Table 3.1 lists the key data elements used in the system.

Data Element	Example	Meaning
Subjects	Mathematics	Subject selected for learning
Difficulty	Easy	Difficulty level of a question
Question	Multiple-choice question	Question presented to the student
Correct Answer	Option B	Correct response used for validation
Score	80	Student's score in a quiz

Table 3.1 Key Data Elements and Their Meanings

The five subjects available in StudyVerse AI, along with their stored question details, are presented in Table 3.2. The question bank contains 75 questions, with 15 questions for each subject, divided equally among Easy, Medium, and Hard difficulty levels.

Department	OPD timing: difficulty levels	Total questions	Topics covered
Mathematics	Easy,Medium,Hard	15	Mathematical concepts and problem solving
Science	Easy,Medium,Hard	15	Scientific concepts
English	Easy,Medium,Hard	15	Grammer and language skills
Computer Science	Easy,Medium,Hard	15	Computing and programming skills
General Knowledge	Easy,Medium,Hard	15	General awareness and facts
Total		75	5 subjects

Table 3.2 Subjects in the Question Bank

Each subject contains five Easy, five Medium, and five Hard questions. The system uses these questions to conduct quizzes, evaluate student performance, and support rule-based adaptive difficulty.

3.4 AI Rules and Decision-Making

StudyVerse AI uses predefined rules to adjust quiz difficulty based on student performance. These rules help the system respond to correct and incorrect answers without requiring machine learning. The optional AI Battle mode uses the Minimax algorithm with Alpha-Beta Pruning to make competitive decisions.

Rule 1: Increase Difficulty

If a student answers two consecutive questions correctly, the system increases the difficulty level.

Rule 2: Decrease Difficulty

If a student answers two consecutive questions incorrectly, the system decreases the difficulty level.

Rule 3: Maintain Difficulty

If the student's performance is mixed, the current difficulty level is maintained.

Rule 4: Heuristic Evaluation

The system evaluates factors such as accuracy, recent performance, response time, current difficulty, streak, and quiz progress to assess student performance.

For example, when a student answers two consecutive questions correctly, the rule-based system increases the difficulty for subsequent questions. In AI Battle mode, Minimax selects a suitable move, while Alpha-Beta Pruning reduces the number of game states evaluated. Thus, predefined rules and classical AI algorithms support adaptive gameplay and decision-making.

3.5 Advantages of the Proposed System

- **Adaptive Difficulty:** Adjusts question difficulty based on students' recent performance using predefined rules.
- **Improved Engagement:** Educational games make learning more interactive and enjoyable.
- **Progress Tracking:** Tracks scores, accuracy, XP, levels, streaks, and achievements.
- **AI-Based Gameplay:** Uses Minimax and Alpha-Beta Pruning in the optional AI Battle mode.
- **Motivation:** Badges, rewards, and levels encourage consistent learning.
- **Performance Evaluation:** Heuristic evaluation considers factors such as accuracy, response time, and recent performance.
- **Centralized Data Management:** Stores student details, questions, quiz results, and progress in a PostgreSQL database.
- **Multiple Subjects:** Supports learning through quizzes in five different subjects.

CHAPTER 4

MODULES DESCRIPTION

StudyVerse AI is divided into several modules that handle student interactions, game sessions, adaptive difficulty, daily challenges, and leaderboard operations. The backend is developed using Node.js and Express.js, with separate controllers and service files to process requests and manage application logic. Authentication middleware protects the routes and restricts access based on user roles.

4.1 LEADERBOARD MODULE

This module displays leaderboard information using the leaderboardController.js controller and leaderboardService.js service. The controller retrieves the authenticated student's ID and passes it, along with the query parameters, to the service. The service processes the request and returns leaderboard data, which is sent to the client in JSON format. Error handling is managed using a try-catch block, with errors forwarded to the next middleware.

4.2 GAME SESSION MODULE

This module manages game session records and student game statistics through gameSessionController.js and gameSessions.js. The createSession() function records a completed game session using the submitted request data. It returns an appropriate HTTP status code depending on whether the session is a duplicate. The getMyStats() function retrieves the authenticated student's game statistics. These endpoints require authentication and the STUDENT role.

4.3 DAILY CHALLENGE MODULE

This module provides the daily challenge through dailyChallengeController.js and dailyChallenge.js. The fetchDailyChallenge() function obtains the authenticated student's ID and calls the service to retrieve the current day's challenge. The result is returned to the frontend in JSON format. Authentication middleware ensures that the endpoint is accessible only to authorized students.

4.4 ADAPTIVE DIFFICULTY MODULE

This module recommends an appropriate quiz difficulty based on student performance. The adaptiveDifficultyController.js controller receives the student's ID and the subject ID from the request query parameters. It then calls getAdaptiveDifficultyRecommendation() in the service layer and returns the recommendation to the frontend. This module supports the platform's rule-based adaptive learning functionality.

4.5 ROUTING AND AUTHENTICATION MODULE

The routing module organizes API endpoints using Express.js routers. The games.js file connects the leaderboard, daily challenge, adaptive difficulty, and game session routes to their respective controllers. Authentication middleware verifies access, while role-based middleware restricts selected endpoints to students. The gameSessions.js router provides endpoints for creating sessions and retrieving student statistics. Errors are forwarded to the application's error-handling middleware for consistent processing.

These modules work together to support StudyVerse AI's game tracking, leaderboard access, daily challenges, and adaptive difficulty recommendations through a structured backend architecture.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

StudyVerse AI was developed using a frontend, backend, and database architecture. The frontend provides the student interface, while the backend processes requests, manages game sessions, and provides adaptive difficulty recommendations. Table 5.1 lists the technologies used in the project.

Component	Details
Frontend	React
Backend	Node.js and express.js
Database	PostgreSQL
ORM	PrismaORM
Authentication	JWT based authentication
Password security	Bycrypt password hashing
API communication	REST API's with Json responses
Adaptive difficulty	Rule based decision making
AI battle mode	Min-Max and Alpha-beta pruning
Development environment	Visual Studios code
Domain	Interactive educational gaming program

Table 5.1 Implementation Environment

5.2 RESULTS

StudyVerse AI provides educational quizzes, game sessions, daily challenges, leaderboard information, and adaptive difficulty recommendations. The backend controllers process requests through service functions and return results to the frontend in JSON format. Table 5.2 summarizes the main functionalities and their expected outputs based on the project design and code structure.

Functionality	System output	Implementation
Student authentication	Login and registration responses	Authentication API's
Game creation	Session creation status	gameSessionController.js
Student statistics	Student game statistics	getMyStats()
Leaderboard	Leaderboard data	leaderboardController.js
Daily Challenge	Current daily challenge	dailyChallengeController.js
Role based access	Access restricted according to user roles	Authentication Middleware

5.2.1 Game Session Management

The game session module records game session data submitted by the student. The backend returns a success response with the session result and uses a different HTTP status code when a duplicate session is detected. Student statistics can also be retrieved through the corresponding API endpoint.

5.2.2 Leaderboard Retrieval

The leaderboard module retrieves leaderboard information by passing the authenticated student's ID and request query parameters to the service layer. The resulting data is returned to the frontend in JSON format for display.

5.2.3 Daily Challenge

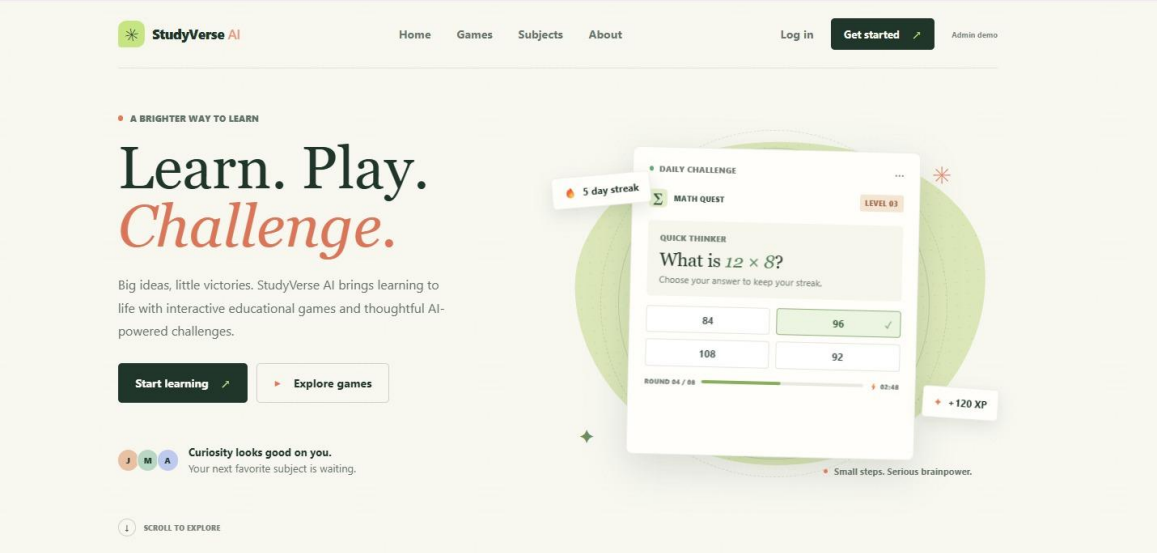
The daily challenge module retrieves the current day's challenge for an authenticated student. The controller calls the corresponding service function and returns the result to the frontend.

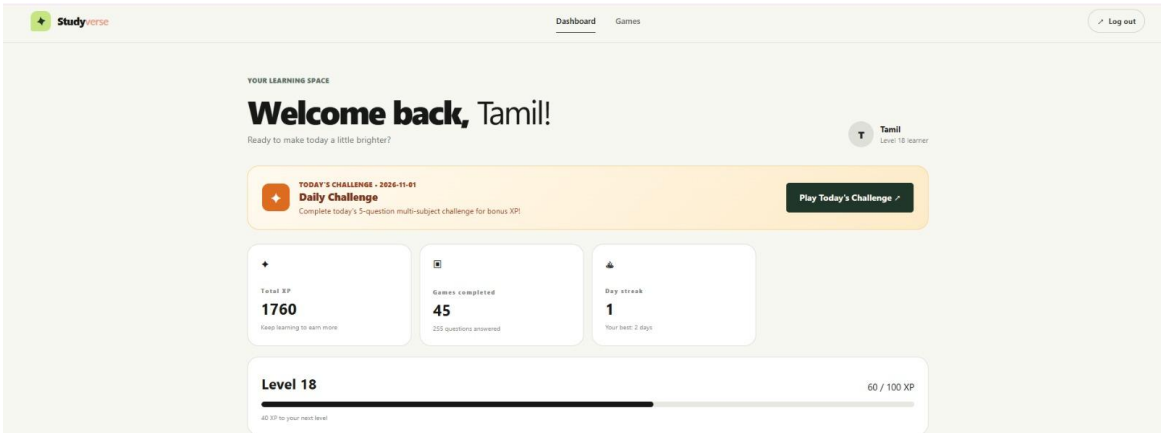
5.2.4 Adaptive Difficulty

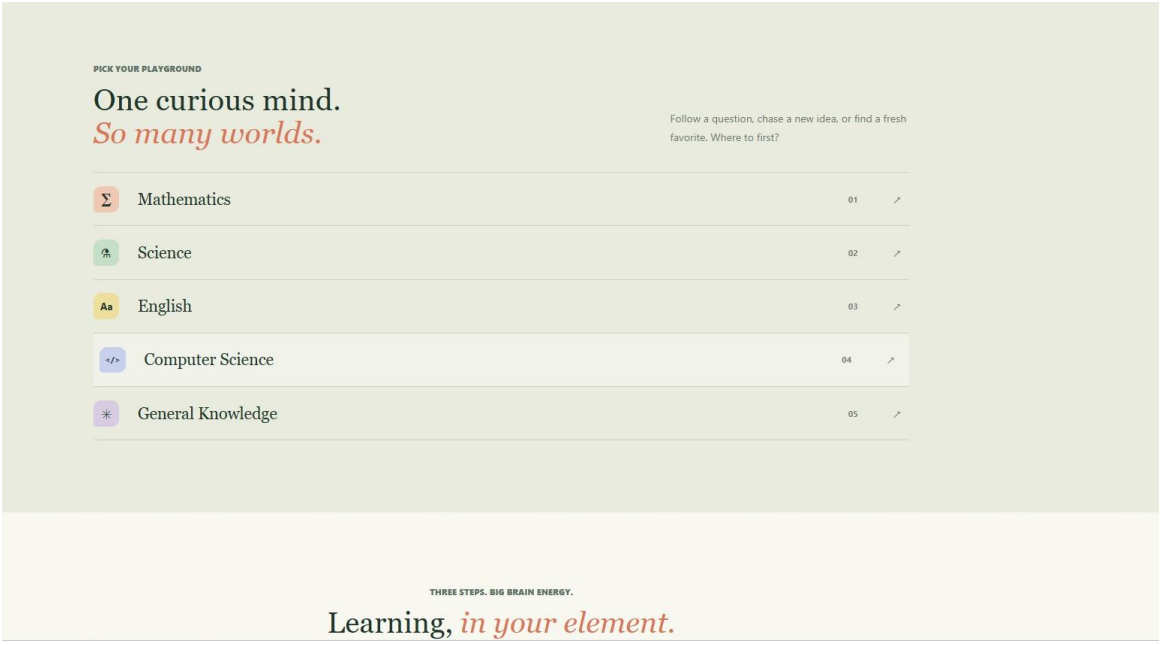
The adaptive difficulty module receives the student's ID and subject ID and requests a difficulty recommendation from the service layer. This functionality supports personalized quiz experiences through rule-based adaptation.

5.2.5 Screenshots

Figures 5.1 to 5.3 should show the main window with an answer and reasoning trace, the administrator login and the administrator panel.




Studyverse
Dashboard
Games
Log out
YOUR LEARNING SPACE
Welcome back, Tamil!
Ready to make today a little brighter?
T
Tamil
Level 18 learner
TODAY'S CHALLENGE · 2026-11-01
Daily Challenge
Complete today's 5-question multi-subject challenge for bonus XP!
Play Today's Challenge ↗
Total XP
1760
Keep learning to earn more
Games completed
45
255 questions answered
Day streak
1
Your best: 2 days
Level 18
60 / 100 XP
40 XP to your next level

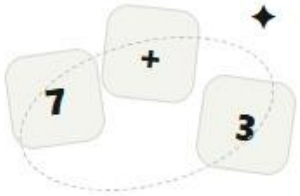

PICK YOUR PLAYGROUND
One curious mind.
So many worlds.
Follow a question, chase a new idea, or find a fresh favorite. Where to first?
Σ
Mathematics
01
↗
Science
02
↗
Aa
English
03
↗
</>
Computer Science
04
↗
General Knowledge
05
↗
THREE STEPS. BIG BRAIN ENERGY.
Learning, in your element.

Pick up your learning journey

Jump into interactive games designed to reinforce concepts through play.

45 games played 255 questions 74.2% avg accuracy

Explore games →



YOUR PRACTICE

Subject progress

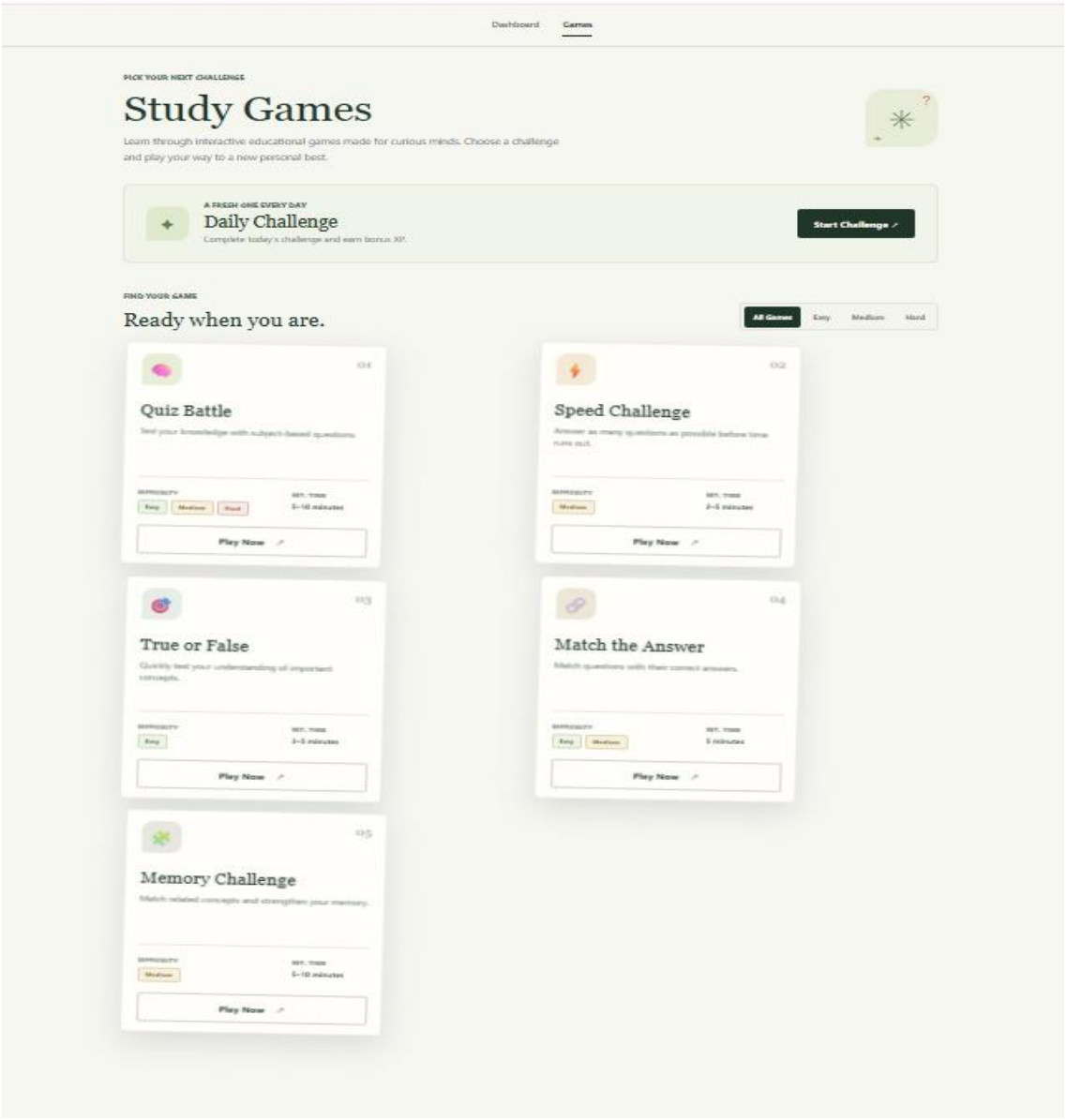
255 questions answered

Science 50 questions • 350 XP		70%
English 44 questions • 200 XP		45%
General Knowledge 36 questions • 190 XP		53%
Mathematics 30 questions • 270 XP		90%
Computer Science 20 questions • 200 XP		100%

RECENT SESSIONS

Learning activity

True or False (Mathematics) 10/13/2026 • 40% accuracy (2/5)	+20 XP
Speed Challenge (Science) 10/13/2026 • 0% accuracy (0/10)	+0 XP
Daily Challenge 10/13/2026 • 0% accuracy (0/5)	+0 XP
Quiz Battle 10/13/2026 • 60% accuracy (3/5)	+30 XP
Quiz Battle 10/13/2026 • 80% accuracy (4/5)	+40 XP


Dashboard
Games
PICK YOUR NEXT CHALLENGE
Study Games
Learn through interactive educational games made for curious minds. Choose a challenge and play your way to a new personal best.
A FRESH ONE EVERY DAY
Daily Challenge
Complete today's challenge and earn bonus XP.
Start Challenge
FIND YOUR GAME
Ready when you are.
All Games
Easy
Medium
Hard
Quiz Battle
Play Now
Speed Challenge
Play Now
True or False
Play Now
Match the Answer
Play Now
Memory Challenge
Play Now

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presents StudyVerse AI, an interactive educational gaming platform that combines quizzes, rule-based adaptive difficulty, and gamification to improve student engagement. Developed using React, Node.js, Express.js, and PostgreSQL, the system supports five subjects and provides features such as game sessions, daily challenges, leaderboards, and progress tracking. Its optional AI Battle mode uses the Minimax algorithm with Alpha-Beta Pruning for decision-making. The platform uses classical AI techniques rather than machine learning, offering an engaging environment for students to practise and monitor their learning progress.

The system's adaptability is limited by its predefined questions and implemented rules. Further improvements can expand its learning features and provide more personalized educational experiences.

6.2 Future Work

- Expand the question bank with more subjects, topics, and difficulty levels.
- Introduce more educational games and interactive learning activities.
- Improve adaptive difficulty using additional student performance indicators.
- Enhance the leaderboard with more detailed performance comparisons.
- Expand progress reports to provide deeper insights into strengths and areas for improvement.
- Improve the AI Battle mode with additional strategies and game scenarios.
- Enhance the administrator dashboard for easier management of questions, users, and challenges.
- Introduce multilingual support to make learning accessible to more students.

CHAPTER 7

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